CLOUD COMPUTING PROJECT

AWS VPC Peering Architecture

Submitted by : Juhi

Date: 22 October 2026

1. Project Overview

The objective of this project is to configure communication between two Amazon Virtual Private Clouds (VPCs) using VPC Peering in Amazon Web Services (AWS). Three EC2 instances are used to test communication within one VPC and between the two VPCs. Connectivity is verified with the ping command.

The project uses VPCs, subnets, route tables, an Internet Gateway where required, security groups, EC2 instances, a VPC Peering connection, and an EC2 Instance Connect Endpoint to access a private instance.

2. Project Objectives

- Create two VPCs with non-overlapping IPv4 CIDR ranges.
- Create subnets and launch EC2 instances in the intended subnets.
- Configure an Internet Gateway and route tables where needed.
- Establish a VPC Peering connection between the two VPCs.
- Configure routes and security groups for the required traffic.
- Test instance connectivity using ICMP ping.
- Access the private EC2 instance through an EC2 Instance Connect Endpoint.

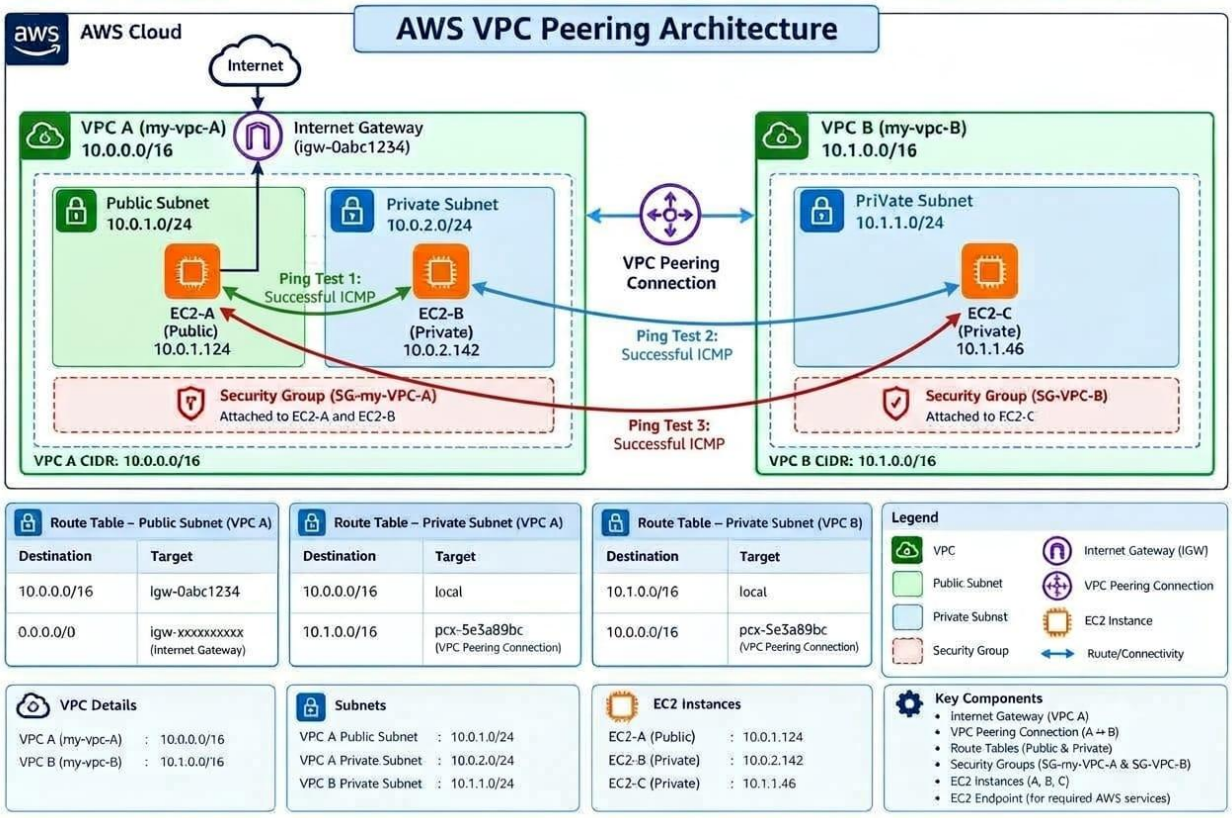
3. Architecture Summary

The architecture contains two VPCs connected through a VPC Peering connection. VPC Peering allows resources to communicate using private IP addresses when the route tables and security groups permit the traffic. Peering does not automatically add routes to route tables.

Instances used during testing:

- EC2 A: private IP 10.0.1.124; used as the first test instance and initial SSH access point.
- EC2 B: private IP 10.0.2.142; a private instance in the first VPC.
- EC2 C: private IP 10.1.1.46; an instance in the second VPC.

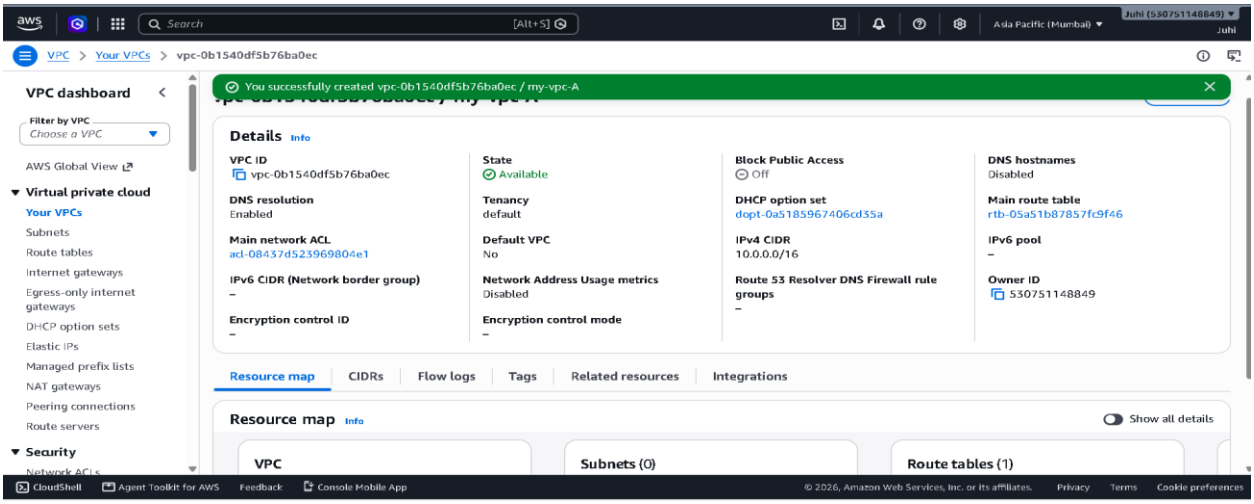
Confirm the actual VPC names, CIDR ranges, subnet names, instance IDs, and route targets in the AWS Console before submitting the report. The addresses above are the private IP addresses used in the tests.

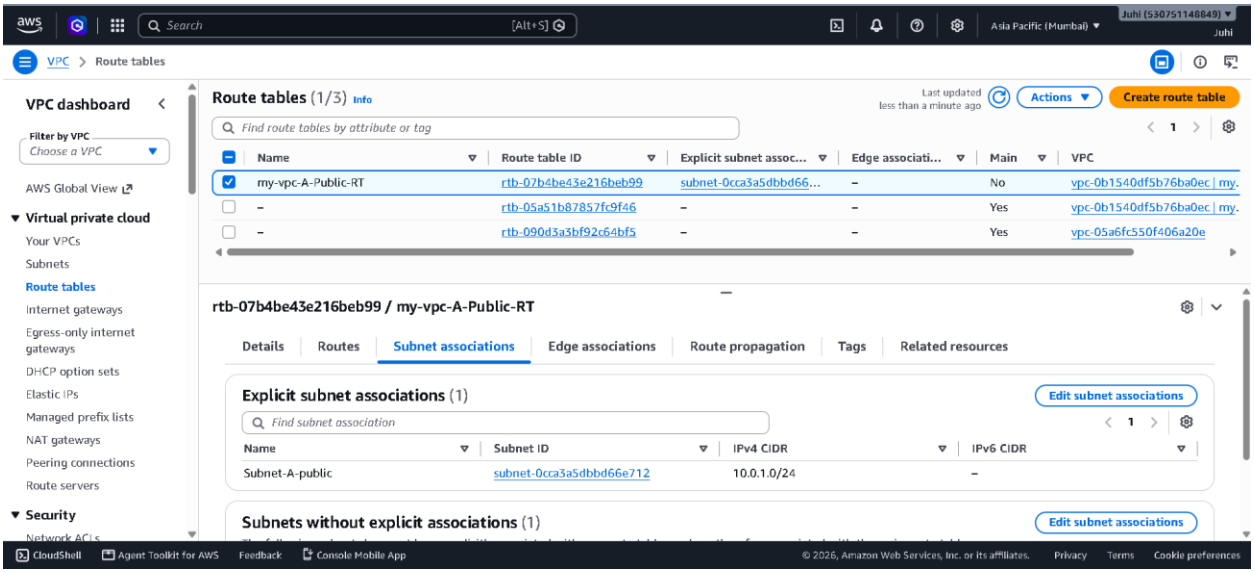


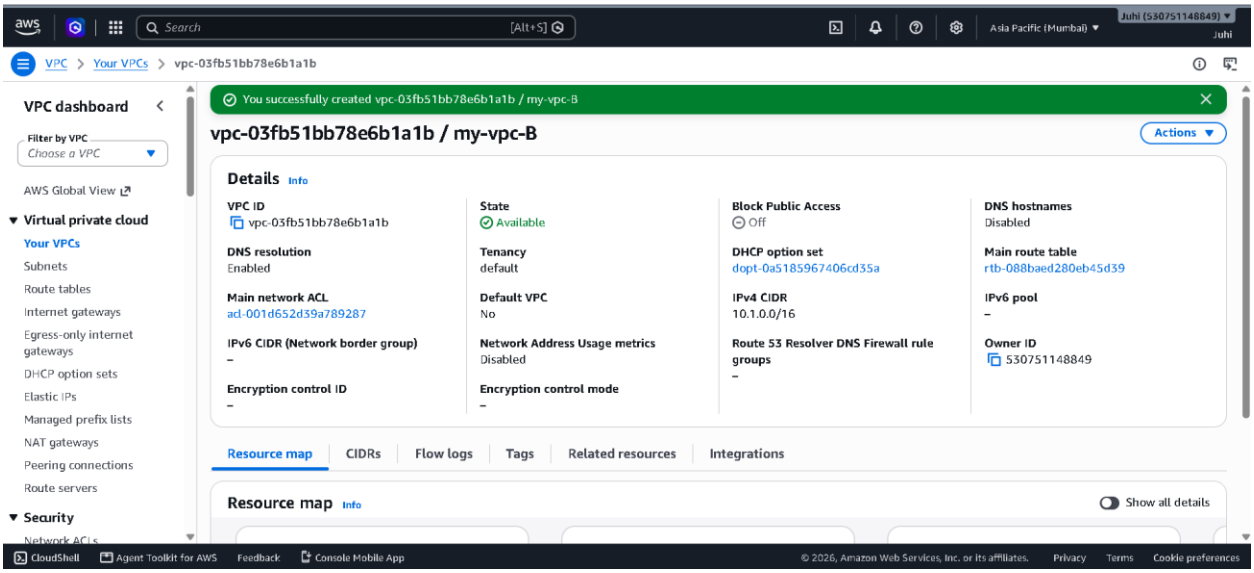
4. Step-by-Step Implementation

Step 1: Create the VPCs

Open the AWS Management Console and navigate to VPC. Create two VPCs with non-overlapping IPv4 CIDR ranges. Non-overlapping ranges allow each VPC to route traffic to the other without ambiguous destinations.

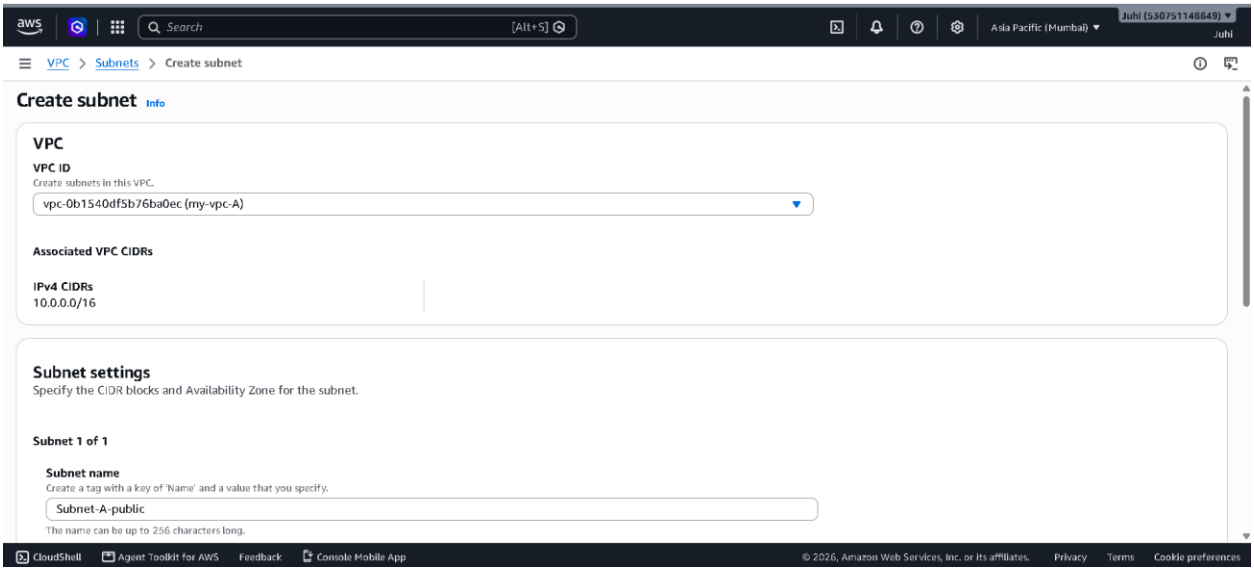


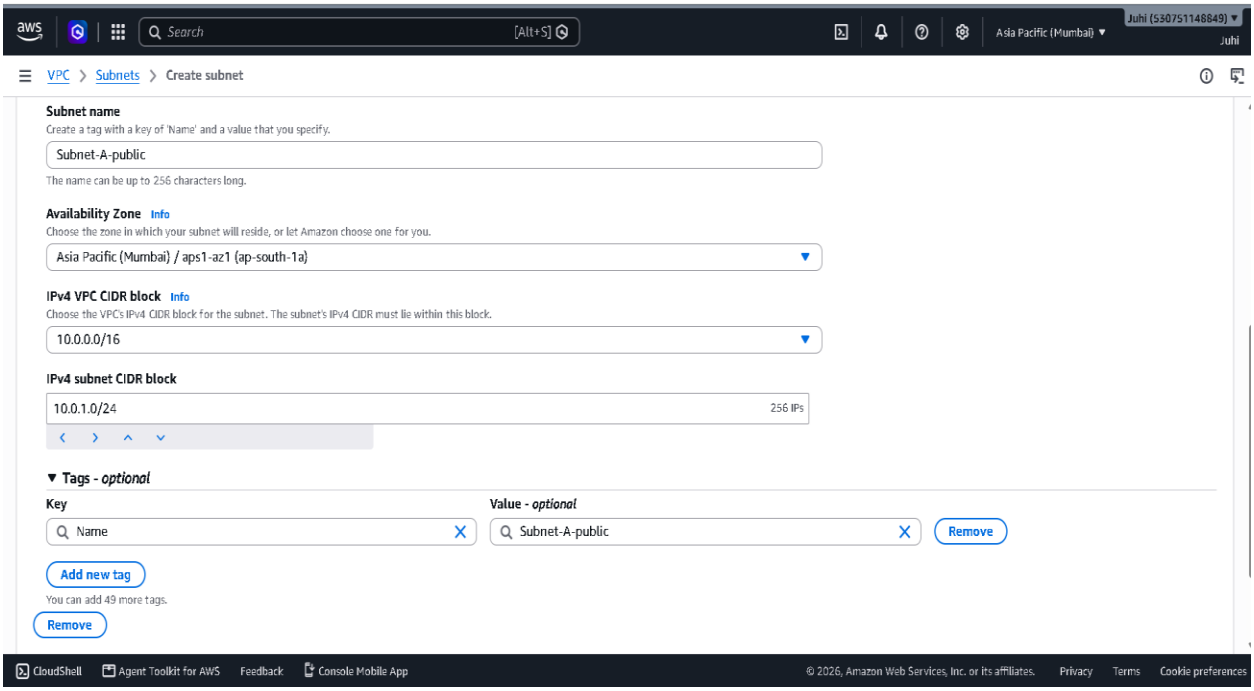


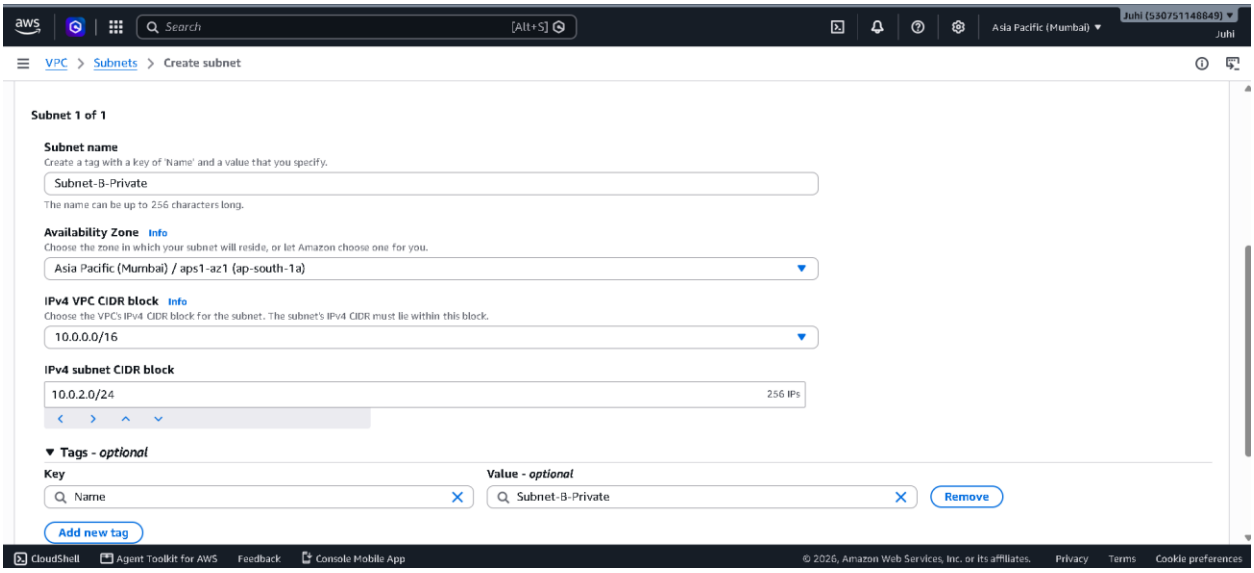


Step 2: Create the Subnets

Create the required subnets inside each VPC and place each EC2 instance in its intended subnet. Based on the tested private IP addresses, EC2 A and EC2 B belong to the 10.0.0.0/16 address range, while EC2 C belongs to the 10.1.0.0/16 address range. Verify the actual CIDRs in the AWS Console.

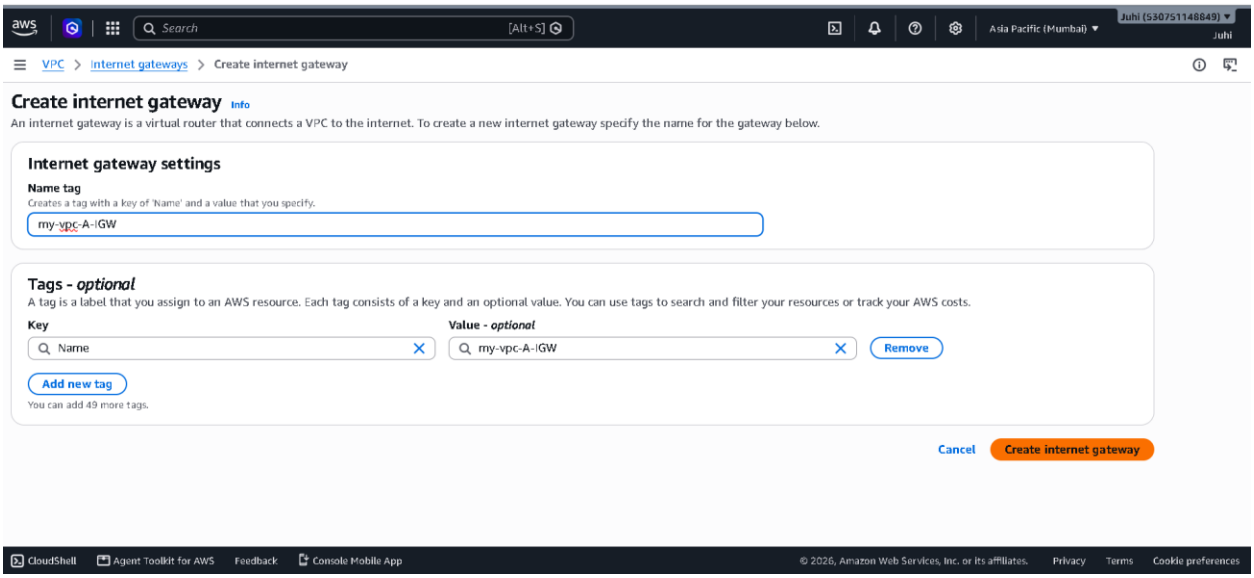

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VPC > Subnets > Create subnet
Create subnet Info
VPC
VPC ID
Create subnets in this VPC.
vpc-0b1540df5b76ba0ec (my-vpc-A)
Associated VPC CIDRs
IPv4 CIDRs
10.0.0.0/16
Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Subnet-A-public
The name can be up to 256 characters long.
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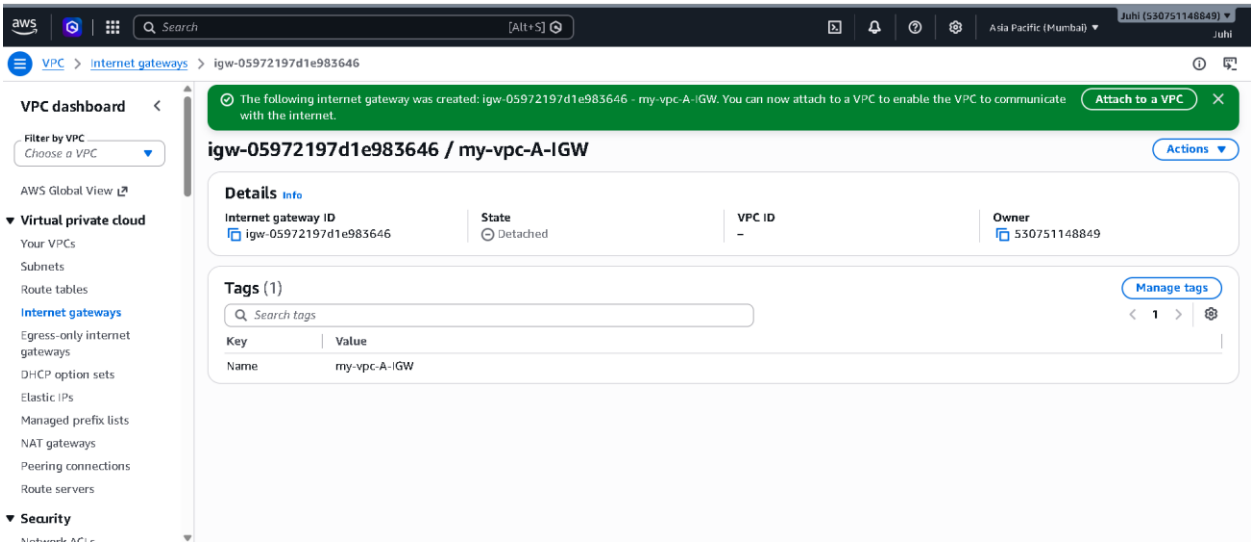

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VPC > Subnets > Create subnet
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Subnet-A-public
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
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IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.1.0/24
256 IPs
▼ Tags - optional
Key
Value - optional
Name
Subnet-A-public
Remove
Add new tag
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Step 3: Configure the Internet Gateway

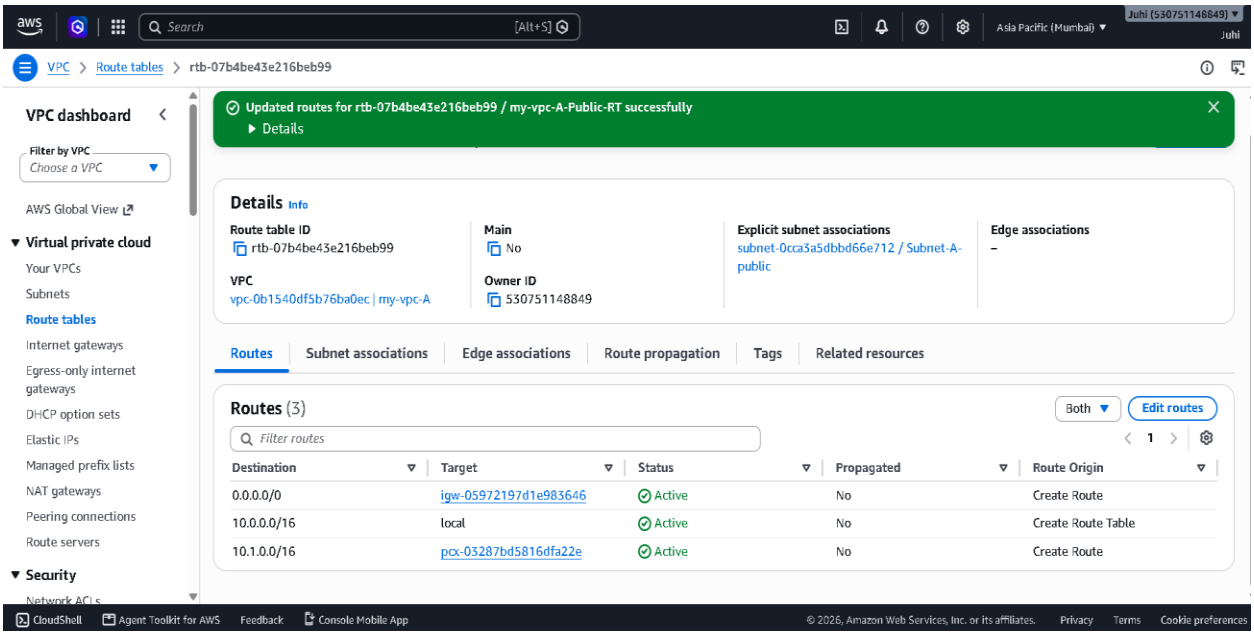
For a VPC that requires internet access, create an Internet Gateway and attach it to the VPC. An Internet Gateway alone does not make an instance publicly reachable. The route table, public IP assignment, and security group rules must also be configured appropriately.

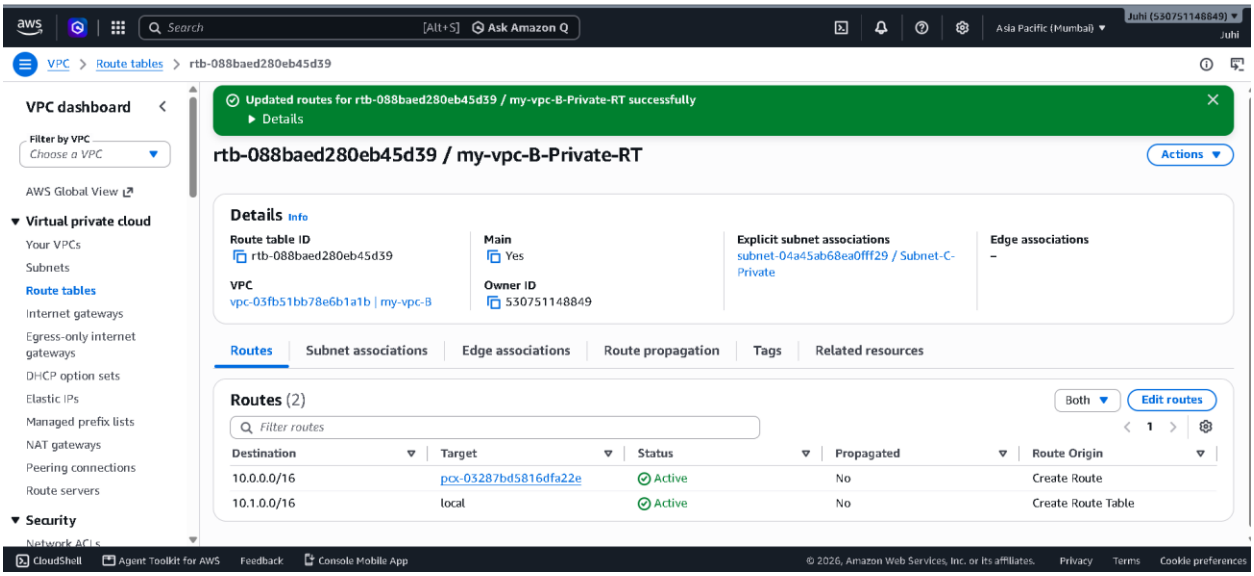




Step 4: Configure Route Tables

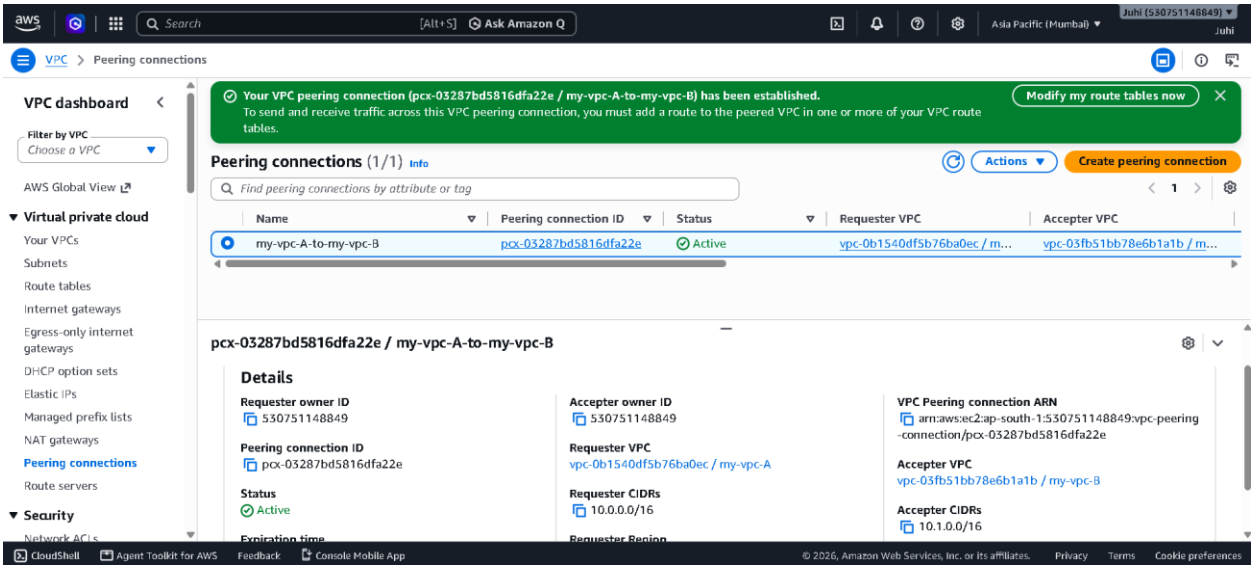
Associate the correct route table with each subnet. Add a route to the remote VPC CIDR through the VPC Peering connection. For example, if the VPC CIDRs are 10.0.0.0/16 and 10.1.0.0/16, the first VPC needs a route to 10.1.0.0/16 through the peering connection, and the second VPC needs a route to 10.0.0.0/16 through the same connection. Retain each VPC's local route.





Step 5: Create the VPC Peering Connection

Create a VPC Peering connection by selecting the requester VPC and accepter VPC. Accept the request and verify that the peering status becomes Active. Add the required routes separately to the relevant route tables.

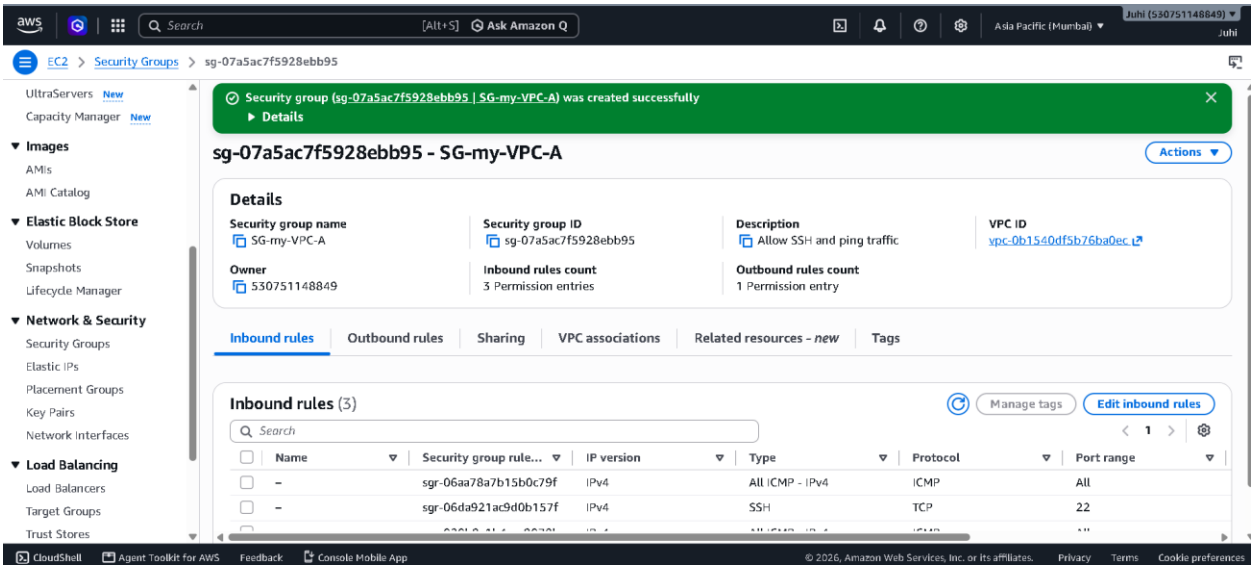


Step 6: Configure Security Groups

Attach suitable security groups to the EC2 instances. Configure only the inbound traffic required for the project. ICMP rules permit ping tests, while SSH uses TCP port 22. For EC2 Instance

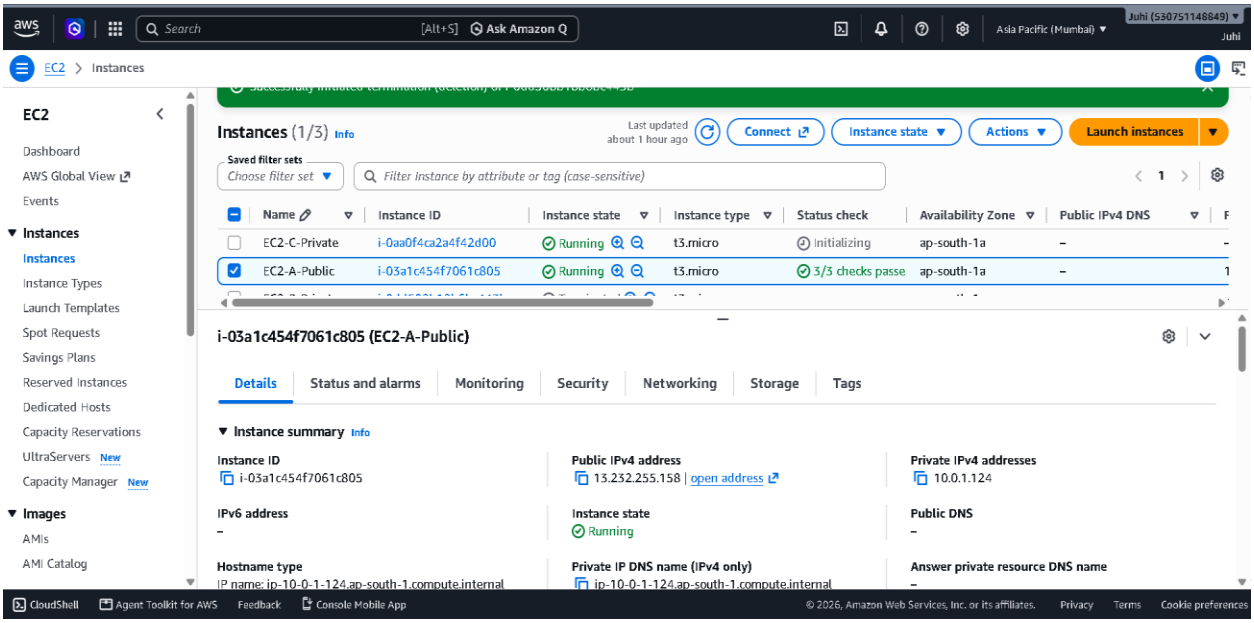
Connect Endpoint access, EC2 B must allow inbound SSH from the endpoint security group, and the endpoint security group must allow outbound SSH to EC2 B. Do not allow SSH from 0.0.0.0/0.

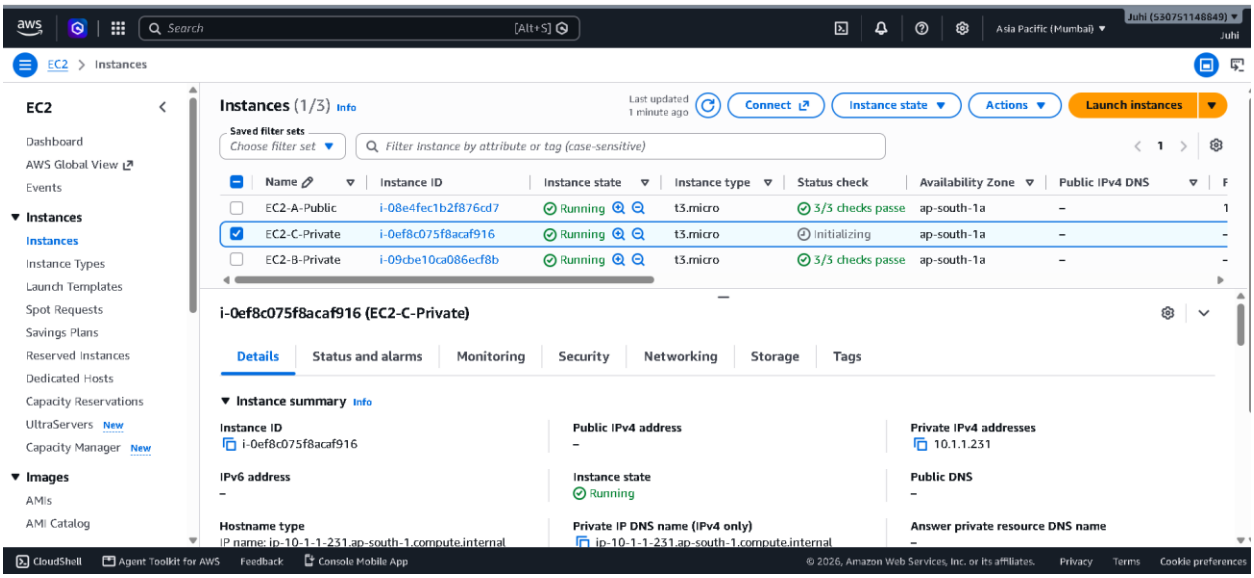
During troubleshooting, SG-my-VPC-A was shown with SSH TCP port 22 allowed from 10.0.1.0/24 and 111.223.30.71/32, plus ICMP rules for 10.0.0.0/16 and 10.1.0.0/16. Review and restrict these rules before final submission, and confirm which instances and endpoints use this security group.

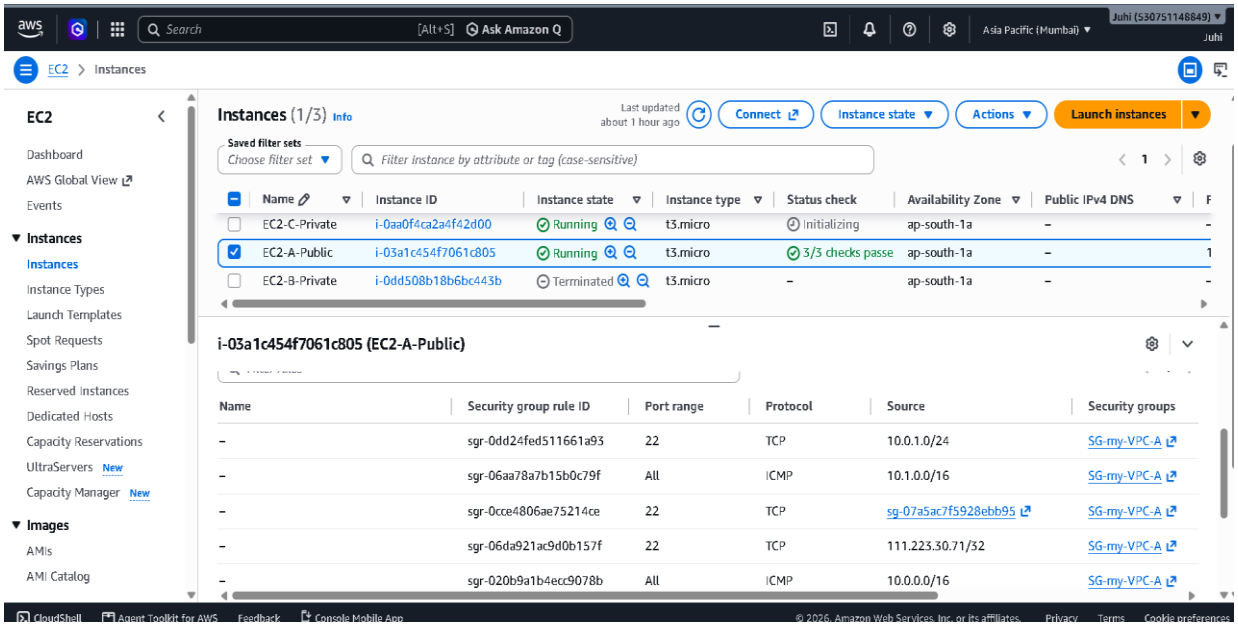


Step 7: Launch and Verify EC2 Instances

Launch the EC2 instances in their intended subnets. Verify each instance's state, private IP address, VPC, subnet, and attached security group. EC2 A was accessed through PuTTY. EC2 B was private and was accessed using an EC2 Instance Connect Endpoint.


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UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instances (1/3) Info
Last updated about 1 hour ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
EC2-C-Private
i-0aa0f4ca2a4f42d00
Running
t3.micro
Initializing
ap-south-1a
EC2-A-Public
i-03a1c454f7061c805
Running
t3.micro
3/3 checks passe
ap-south-1a
i-03a1c454f7061c805 (EC2-A-Public)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-03a1c454f7061c805
Public IPv4 address
13.232.255.158 | open address
Private IPv4 addresses
10.0.1.124
IPv6 address
Instance state
Running
Public DNS
Hostname type
Private IP DNS name (IPv4 only)
Answer private resource DNS name
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Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instances (1/3) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
EC2-A-Public
i-08e4fec1b2f876cd7
Running
t3.micro
3/3 checks passe
ap-south-1a
EC2-C-Private
i-0ef8c075f8acaf916
Running
t3.micro
Initializing
ap-south-1a
EC2-B-Private
i-09cbe10ca086ecf8b
Running
t3.micro
3/3 checks passe
ap-south-1a
i-0ef8c075f8acaf916 (EC2-C-Private)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0ef8c075f8acaf916
Public IPv4 address
Private IPv4 addresses
10.1.1.231
IPv6 address
Instance state
Running
Public DNS
Hostname type
Private IP DNS name (IPv4 only)
Answer private resource DNS name
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Step 8: Test Connectivity

From the EC2 A terminal, run the following commands to test connectivity to EC2 B and EC2 C:

```
ping -c 4 10.0.2.142
```

```
ping -c 4 10.1.1.46
```

Then, from the EC2 B terminal, run:

```
ping -c 4 10.1.1.46
```

All three tests completed successfully and ping packets were received. This confirms ICMP connectivity for EC2 A to EC2 B, EC2 A to EC2 C, and EC2 B to EC2 C.

```
ubuntu@ip-10-0-1-124:~$ ping -c 4 10.0.2.142
PING 10.0.2.142 (10.0.2.142) 56(84) bytes of data.
64 bytes from 10.0.2.142: icmp_seq=1 ttl=64 time=0.460 ms
64 bytes from 10.0.2.142: icmp_seq=2 ttl=64 time=0.174 ms
64 bytes from 10.0.2.142: icmp_seq=3 ttl=64 time=0.183 ms
64 bytes from 10.0.2.142: icmp_seq=4 ttl=64 time=0.165 ms
```

```
ubuntu@ip-10-0-1-124:~$ ping -c 4 10.1.1.46
PING 10.1.1.46 (10.1.1.46) 56(84) bytes of data.
64 bytes from 10.1.1.46: icmp_seq=1 ttl=64 time=0.800 ms
64 bytes from 10.1.1.46: icmp_seq=2 ttl=64 time=0.180 ms
64 bytes from 10.1.1.46: icmp_seq=3 ttl=64 time=0.187 ms
64 bytes from 10.1.1.46: icmp_seq=4 ttl=64 time=0.194 ms

--- 10.1.1.46 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3090ms
rtt min/avg/max/mdev = 0.180/0.340/0.800/0.265 ms
ubuntu@ip-10-0-1-124:~$
```

```
ubuntu@ip-10-0-2-142:~$ ping -c 4 10.1.1.46
PING 10.1.1.46 (10.1.1.46) 56(84) bytes of data.
64 bytes from 10.1.1.46: icmp_seq=1 ttl=64 time=0.606 ms
64 bytes from 10.1.1.46: icmp_seq=2 ttl=64 time=0.226 ms
64 bytes from 10.1.1.46: icmp_seq=3 ttl=64 time=0.165 ms
64 bytes from 10.1.1.46: icmp_seq=4 ttl=64 time=0.198 ms

--- 10.1.1.46 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3102ms
rtt min/avg/max/mdev = 0.165/0.298/0.606/0.178 ms
ubuntu@ip-10-0-2-142:~$
```

5. Troubleshooting Performed

- Initial PuTTY/SSH access failed, so the connection settings and access path were checked.
- EC2 B was private and could not be accessed directly from the local computer using a public IP.
- An EC2 Instance Connect Endpoint was created to access the private instance.
- The endpoint initially showed Pending and later changed to Available.
- After connection and security settings were addressed, EC2 B terminal access succeeded.
- All three intended ping tests succeeded.

A successful ping verifies ICMP connectivity only. Other protocols, such as SSH, HTTP, or application-specific ports, require their own security group rules and service checks.

6. Results

The project demonstrated private-IP communication between EC2 instances in the same VPC and between instances in two VPCs connected by VPC Peering. Successful ICMP tests indicate that routing and security group settings permit the tested traffic paths.

7. Security and Cost Considerations

- Allow only the inbound ports and source addresses required for the project.
- Do not expose SSH to all IPv4 addresses (0.0.0.0/0).
- Delete the EC2 Instance Connect Endpoint when it is no longer needed, after checking any project requirements.
- Stop or terminate EC2 instances when they are no longer needed, depending on whether the project must be preserved.
- Review AWS Billing and Cost Explorer for chargeable resources. A VPC itself generally has no hourly charge, but associated resources may incur costs.
- Keep the VPCs for future practice if useful; deleting them is not required for project completion.

8. Conclusion

The AWS VPC Peering project was implemented and tested successfully. Two VPCs were connected using VPC Peering, route tables were configured for remote CIDR ranges, and security groups permitted the required traffic. Three successful ping tests demonstrated connectivity between EC2 A, EC2 B, and EC2 C along the tested paths. Insert the relevant AWS Console and terminal screenshots in the marked locations to complete the report.

9. Screenshot Submission Checklist

- VPC list and CIDR ranges
- Subnet list and VPC associations
- Internet Gateway attachment
- Route tables for both VPCs
- VPC Peering connection showing Active
- Security group rules for EC2 instances
- EC2 instance list and private IP addresses
- EC2 Instance Connect Endpoint showing Available
- Ping output for EC2 A to EC2 B
- Ping output for EC2 A to EC2 C
- Ping output for EC2 B to EC2 C